

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Weightage:	20% of CIA
Total Marks:	20 (2 Questions × 10 Marks)	Date:	29-07-2026
Student Name:	A. Archishman	Reg. No.:	192425341

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

(a) Interpretation of the Scenario and Type of Noise

The repetitive horizontal stripe pattern observed across the captured images is a classic signature of periodic noise. Periodic noise is generated when the imaging or transmission system is disturbed by electrical or electromechanical interference that repeats at a fixed frequency — for example, interference from AC power lines, motors, or other electronic equipment operating near the camera or sensor circuitry. Because the interference is periodic in nature, it appears in the image as regularly spaced sinusoidal stripe patterns rather than as random speckles, which distinguishes it clearly from noise types such as Gaussian or salt-and-pepper noise.

(b) Effect of the Degradation on Inspection Accuracy

In an automated quality inspection pipeline, the system relies on detecting fine variations in pixel intensity to identify surface defects such as cracks, scratches, dents, or discoloration on metal components. The periodic stripe interference superimposes a strong, regular intensity variation across the entire image. This has several harmful effects:

- It masks or distorts genuine defect edges, causing real defects to be missed (false negatives).
- It can be mistaken for a defect pattern itself, triggering false alarms (false positives).
- It reduces the effective contrast and sharpness of the image, lowering the reliability of any subsequent edge-detection or segmentation stage.

Overall, the presence of periodic noise directly undermines the accuracy, consistency, and trustworthiness of the automated inspection system.

(c) Recommended Enhancement Technique

The most appropriate technique to remove this type of interference is frequency-domain filtering using the Fourier Transform, specifically Notch Filtering. The image is first transformed into the frequency domain using the 2-D Discrete Fourier Transform (DFT / FFT). Because periodic noise appears as a small number of concentrated, symmetric bright spikes at specific frequency locations in the Fourier spectrum (away from the origin), these spikes can be precisely located and suppressed. A notch reject filter (ideal, Butterworth, or Gaussian notch filter) is then designed to attenuate only the frequency components at and around these spike locations while preserving all other frequencies. The inverse Fourier Transform is finally applied to reconstruct a clean spatial-domain image with the stripe interference removed.

(d) Justification for Preferring the Frequency-Domain Approach

Frequency-domain notch filtering is more suitable than spatial-domain methods (such as averaging or median filtering) for the following reasons:

- Precise, targeted removal: Periodic noise occupies a very specific, narrow location in the frequency spectrum. Notch filtering can surgically remove exactly this component without touching the rest of the image content.
- Preservation of image detail: Spatial filters such as averaging or median filters act uniformly across the whole image and would blur genuine defect edges and fine surface texture along with the noise, which is unacceptable in quality inspection.
- Effectiveness against structured, repetitive patterns: Spatial filters are designed mainly for random noise and are inefficient at removing structured periodic patterns, which instead concentrate very clearly in the frequency domain.
- Better trade-off between noise removal and information retention: Since the notch filter targets only the offending frequencies, the fine details required to detect real defects are retained with minimal signal loss.

(e) Logical Sequence of the Solution

- Step 1: Acquire the noisy image showing repetitive horizontal stripe interference.
- Step 2: Apply the 2-D Fourier Transform to convert the image into the frequency domain.
- Step 3: Examine the Fourier spectrum and identify the symmetric bright spikes corresponding to the periodic noise.
- Step 4: Design and apply a notch reject filter centered at the identified spike locations.
- Step 5: Apply the inverse Fourier Transform to obtain the restored, noise-free spatial-domain image.
- Step 6: Use the enhanced image for reliable defect detection and quality inspection.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

(a) Interpretation of the Scenario and the Challenge Involved

Heavy rain introduces random noise and visual clutter into the images captured by the vehicle's camera — raindrops, water spray, reduced visibility, and sensor-level random variations all degrade image quality. The central challenge is that the noise must be suppressed to obtain a clean image, but this cannot come at the cost of blurring the lane markings and road edges, since these edges are the critical features the autonomous navigation system depends on for lane-keeping, path planning, and collision avoidance. A conventional smoothing filter would remove noise but would also blur these essential edges, which could compromise the safety of the vehicle.

(b) Importance of Preserving Edges While Removing Noise

Lane markings and road boundaries are typically represented in the image as sharp intensity discontinuities (edges). If an enhancement technique smooths these edges along with the noise, the navigation system may:

- Lose accurate localization of lane boundaries, causing incorrect steering decisions.
- Fail to distinguish the road edge from the surrounding environment in adverse weather.
- Experience delayed or inaccurate object and obstacle detection.

Therefore, edge preservation is not just a quality concern but a direct safety requirement in autonomous driving systems.

(c) Recommended Filtering Technique

The most suitable technique for this scenario is Bilateral Filtering. Bilateral filtering is a non-linear, edge-preserving spatial filter that smooths an image by combining two weighting functions: one based on spatial distance between pixels (as in a standard Gaussian filter) and another based on the difference in intensity values between the pixels (a range/intensity kernel). A pixel is smoothed only using neighboring pixels that are both spatially close and have similar intensity values.

(d) Justification for the Chosen Filter

Bilateral filtering preserves important image details for the following reasons:

- **Selective smoothing:** Because the range (intensity) kernel gives low weight to neighboring pixels with very different intensities, pixels across a strong edge (like a lane marking boundary) are not averaged together, so the edge remains sharp.
- **Effective noise suppression in flat regions:** In regions without strong intensity changes (such as the road surface), pixels have similar intensities, so the range kernel allows normal Gaussian-like averaging, effectively removing rain-induced random noise.
- **Better than linear smoothing filters:** Unlike an average or standard Gaussian filter, which blurs edges uniformly, bilateral filtering adapts its smoothing behavior locally based on image content, making it ideal for scenarios that demand both denoising and edge sharpness simultaneously.
- **Safety-critical suitability:** The preserved sharpness of lane markings and road edges ensures that downstream lane-detection and navigation algorithms continue to receive reliable, high-fidelity edge information even in noisy rainy conditions.

(e) Systematic Explanation of the Solution

- **Step 1:** Acquire the road image captured by the vehicle camera during heavy rain.
- **Step 2:** Recognize the dual requirement — remove random rain-induced noise while preserving edges.
- **Step 3:** Apply bilateral filtering, computing pixel weights from both spatial proximity and intensity similarity.
- **Step 4:** Smooth noise in homogeneous regions (road surface, sky) while retaining sharp transitions at lane markings and road boundaries.
- **Step 5:** Pass the enhanced, edge-preserved image to the lane-detection and navigation module for safe, accurate driving decisions.